# Poverty and Zakat Distribution Analysis and Forecasting in Malaysia

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***Abstract*—Poverty is one of the important socio-economic problems in Malaysia that can have a negative impact on the welfare of communities and on access to basic needs. The project's goal is to examine poverty trends and the distribution of zakat in various states within Malaysia from 2007 to 2022. The data utilized for this project came from official government sources. Data preprocessing techniques such as data cleaning, formatting, and merging were performed using Python in Google Colab. The patterns of poverty and zakat distribution across states and years were visualized through descriptive analysis methods such as bar charts, line graphs and scatter plots. Moreover, predictive analysis was performed in the form of Multiple Linear Regression and Decision Tree Regression models to estimate future trends of poverty based on historical poverty and distribution of zakat data. The performance of both models was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R-squared (R2). The findings can assist policymakers, welfare organizations, and zakat institutions in making informed decisions related to poverty alleviation, welfare planning, and zakat allocation.**

***Keywords—Poverty Analysis, Poverty Forecasting, Zakat Distribution, Predictive Analytics, Multiple Linear Regression, Decision Tree Regression, Malaysia***

## I. INTRODUCTION

Today, poverty remains one of the major socio-economic challenges in Malaysia, affecting access to basic necessities such as food, healthcare, education, and housing. Different states might have different levels of poverty due to differences in economic development, employment opportunities, and living expenses. Within the Malaysian Muslim community, Zakat serves as an important Islamic social finance mechanism for assisting vulnerable groups, particularly the Asnaf, through financial aid and welfare programmes. Descriptive and predictive analytics can be used to better understand poverty trends and support more efficient welfare planning in Malaysia. Therefore, the purpose of this research is to analyze poverty trends and Zakat distribution in Malaysia to provide insight that can further improve community welfare as an effort to reduce poverty.

Although poverty reduction programmes and Zakat distribution have been implemented throughout Malaysia, welfare institutions and policymakers often rely on historical records and static estimations, which may fail to adequately anticipate future poverty trends and lead to the inefficient allocation of financial assistance. Therefore, to tackle these issues, there is a need for a data-driven approach that can analyse historical poverty patterns and forecast future poverty levels to support more proactive decision-making.

The research problem can be classified as a prediction and forecasting problem. The study aims to find the poverty trends pattern using existing data and utilize it to estimate future poverty levels by using year and Zakat distribution information as predictor variables. Machine learning techniques such as Multiple Linear Regression and Decision Tree Regression are employed to identify patterns within the data and generate forecasts that can assist stakeholders in anticipating future welfare requirements.

The problem exists within the Business Analytics domain, especially in the areas of Islamic social finance, social welfare analytics, and resource management. From a business perspective, government agencies, Zakat institutions, and welfare organizations must effectively manage limited resources in order to optimize the effectiveness of initiatives aimed at reducing poverty. (World Bank, 2023).

In the context of Islamic social finance and welfare management, Zakat institutions and policymakers need reliable information about future poverty trends to ensure that financial assistance is distributed effectively.This

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research applies descriptive analytics and predictive analytics using machine learning techniques to analyse poverty rates and Zakat distribution across Malaysian states from 2007 to 2022. The resulting application can be viewed as a predictive decision support system, which enables stakeholders to forecast future poverty levels, optimize the allocation of Zakat funds, and estimate welfare resource requirements. Through proactive planning and evidence-based decision-making, the data product can contribute to improved community welfare and support Malaysia's progress towards achieving Sustainable Development Goal 1 (No Poverty)(Dpicampaigns, 2025).

## II. BACKGROUND

All around the world, including Malaysia are facing one of the most significant socio-economic challenges, which is poverty. Basic needs such as housing, food, healthcare, and education are limited making them a luxury in some countries. It also raises inequality and lowers economic opportunities. Despite numerous initiatives made by the government, poverty levels still vary among states due to variations in economic structures across the states.

Zakat, an Islamic finance mechanism which is aimed to help populations in need particularly Asnaf, is implemented throughout Malaysia. Through systematic collection and distribution of Zakat funds, Islamic religious institutions provide financial support, educational assistance, healthcare aid, and other welfare programs to eligible recipients. In order to assist initiatives in reducing poverty and enhancing community welfare, administering Zakat is important.

Applying data analytics in identifying poverty trends and the impact of welfare efforts has been made possible with the availability of socio-economic data. Descriptive analytics can visualize the historical pattern, regional variations, and correlations between poverty rates and Zakat distribution. Predictive analytics helps researchers and policymakers to predict future poverty trends by using the historical observations.

Hidden patterns in the dataset are able to be identified through the usage of machine learning techniques, as they have become more powerful and useful in forecasting socio-economic indicators. While Decision Tree Regression uses hierarchical decision rules to capture non-linear correlations, Multiple Linear Regression is frequently employed to represent relationships between dependent and independent variables. Through the combination of descriptive and predictive analytics, organizations are able to create more solutions reducing poverty.

This study focuses on examining the distribution of Zakat and poverty rates across several states in Malaysia. Through data visualization and forecasting techniques. The study intends to offer insights that will help politicians, welfare organizations, and Zakat institutions design more successful community welfare and poverty reduction initiatives

## III. PROBLEM STATEMENT

Poverty continues to be one of the major socio-economic issues despite the relentless efforts of the Malaysian government and welfare institutions in addressing the issue of poverty, impacting on the well-being and quality of life of many individuals and households. The economic growth, employment opportunities and resource availability are different among the states of Malaysia, resulting in disparities in poverty rates, which in turn show differences in living standard or welfare conditions. Besides, zakat is an essential instrument of the welfare system which can assist economically vulnerable communities. But it is important to understand the link between poverty rates and the distribution of zakat to enable welfare to be distributed optimally and to the areas that need it more.

Most of the existing reports on poverty are descriptive, which offer information on the state of poverty in the past and present but do not provide any information on the future trends of poverty. This can pose obstacles for policy makers, welfare organisations, and zakat institutions in their planning of proactive poverty alleviation strategies and resources provisioning. Thus, there is a need for a data-driven predictive approach that can analyse historical poverty and the distribution of zakat data in order to establish patterns and forecast poverty rates. This study will use machine learning techniques like Multiple Linear Regression and Decision Tree Regression to assist decision making in poverty reduction, welfare planning and management of zakat distribution in Malaysia.

## IV. RESEARCH QUESTION

The below-mentioned research questions have been proposed to facilitate the research process in the investigation of the above-mentioned aspects along with the applicability of machine learning models in predicting poverty rates in Malaysia in relation to SDG 1: No Poverty:

1) What are the patterns of poverty rates for Malaysian states between 2007 to 2022?

2) What are the patterns of zakat distribution for Malaysian states between 2007 to 2022?

3) How are poverty rates and zakat distribution related to each other in Malaysia?

4) Can machine learning algorithms predict poverty rates based on historical poverty rates and zakat distributions?

5) Which algorithm produces better results: Multiple Linear Regression or Decision Tree Regression?

## V. HYPOTHESIS

Based on the research questions stated above, hypotheses are generated to statistically investigate the relationships that exist between poverty, zakat allocation, and machine learning's predictive power in Malaysia:

1) There is a significant association between the level of poverty rates and zakat distribution among states in Malaysia.

2) The inclusion of historical data on poverty rate and zakat distributions has significantly enhanced the predictability of poverty levels.

3) There is a significant distinction in prediction performances between Multiple Linear Regression and Decision Tree Regression.

## VI. RESEARCH OBJECTIVES

The research objective of these project are as follow;

1) To analyse the trends, patterns, and relationships of poverty rates and Zakat distribution across Malaysian states from 2007 to 2022 using descriptive analytics, predictive analytics and data visualization.

2) To develop and compare Multiple Linear Regression and Decision Tree Regression models for forecasting poverty levels based on year and Zakat distribution data.

3) To evaluate the performance of the forecasting models using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and coefficient of determination ($R^2$).

4) To provide a predictive decision support framework that can assist policymakers, welfare organizations, and Islamic social finance institutions in improving welfare planning and optimizing Zakat allocation.

5) To support Malaysia's efforts towards achieving Sustainable Development Goal 1 (SDG 1: No Poverty) through the application of descriptive and predictive analytics.

## VII. RESEARCH SIGNIFICANCE

This research provides a data-driven understanding of poverty trends and Zakat distribution in Malaysia to support policymakers, charity organizations, Islamic financial institutions, and academic researchers.

Firstly, the study helps policymakers and government organizations to understand the various trends of poverty in the states of Malaysia. To ensure the resources reach those highly in-need, the identification of poverty patterns can aid in the development of more focused and evidence-based policies aimed at reducing poverty.

Secondly, by helping Zakat institutions to understand the current trends between poverty labels and Zakat distribution, planning and allocation of Zakat funds becomes better by thoroughly identifying those truly in-need. The forecasting models created in this work may also help organizations prepare proactive welfare policies and anticipate future resource needs.

Thirdly, by encouraging the creation of long-term programs to combat poverty, the study advances communal welfare. The future trends of poverty will be able to be predicted by organizations accurately before it happens. This permits the implementation of early intervention strategies.

This study demonstrates how machine learning and descriptive analytics may be used to address socioeconomic problems from an academic perspective. Predictive analytics can be used in the domains of Islamic social finance, public policy, and social welfare through the application of Decision Tree Regression and Multiple Linear Regression.

Through the usage of the available information, it contributes to the larger goal of accomplishing Sustainable Development Goal 1 (SDG 1: No Poverty) by helping future efforts in reducing poverty and enhancing the efficacy of welfare services across Malaysia.

## VIII. LITERATURE REVIEW

Several researches have been performed on issues related to poverty measures, vulnerability measures, and statistical as well as machine learning approaches for predicting and classifying poverty. The key aim of this research is highly consistent with the overall goal of the project being carried out, especially when it comes to better knowledge of poverty identification and forecasting approaches. The table presented below highlights this research and their authors.

TABLE 1: LITERATURE REVIEWS

| Literature Reviews | | | | |
|---|---|---|---|---|
| **Year** | **Authors** | **Research Problem** | **Main Technique Applied** | **Results** |
| 2020 | Bakar et al. | Poverty classification using multidimensional socio-economic indicators in Malaysia. | Decision Tree, Random Forest, Linear Regression | Random Forest outperformed Decision Tree and Linear Regression in classifying poverty using eKasih dataset. |
| 2020 | Zamzuri et al. | Forecasting poverty using machine learning classification methods. | Random Forest, XGBoost | Ensemble models improved poverty classification accuracy using socio-economic variables. |
| 2021 | Sohnesen & Stender | Evaluating machine learning models for poverty prediction across developing regions. | Random Forest Regression | Random Forest provides higher predictive accuracy compared to traditional regression models in poverty estimation. |
| 2020 | Ishak & Jaapar | Forecasting Zakat collection trends for improved welfare planning. | Holt-Winters Time Series Forecasting | Demonstrated that time-series forecasting can effectively predict Zakat collection patterns for better resource allocation. |
| 2021 | Yusof et al. | Predicting Zakat collection using artificial intelligence methods. | LSTM Deep Learning Model | Showed that deep learning models such as LSTM improve accuracy in forecasting Zakat revenue trends. |
| 2023 | Saharudin et al. | Relationship between macroeconomic factors and poverty levels in Malaysia. | Nonlinear ARDL Model | Identified significant long-term relationships between economic indicators and poverty trends. |
| 2023 | Nurfazrina et al. | Automating poverty classification using structured socio-economic datasets. | Ensemble Machine Learning (Random Forest, XGBoost) | Found that ensemble models improve poverty classification accuracy using multidimensional socio-economic variables. |
| 2024 | Ngong'ho Bujiku Sende et al. | Predicting impoverished households using hybrid | Hybrid Machine Learning (K-Prototypes Clustering | Proposed a hybrid model that combines clustering and |

| | | | | |
|---|---|---|---|---|
| | | machine learning to reduce human bias. | + Decision Tree) | classification for automated poverty prediction. |
| 2020 | Aulia et al. | Predicting regional poverty using alternative digital footprint data. | K-Nearest Neighbor (KNN) with Feature Selection | Demonstrated that non-traditional datasets such as digital footprints can effectively support poverty classification. |
| 2024 | Ramzan et al. | Poverty prediction using deep learning feature fusion | CNN (ResNet50 + Gated Attention Module) | Improved poverty prediction accuracy using advanced feature fusion deep learning architecture |

The application of artificial intelligence in classifying poverty by using different methods, such as Decision Tree, Random Forest, and Linear Regression, was studied by Bakar et al. (2020) with regards to poverty classification in Malaysia with the aid of multidimensional socio-economic indicators. According to their results, Random Forest outperformed the other two methods because of its ability to model complex inter-relationships among socio-economic factors. Through this study, one can see how data-driven methods can be used in identifying communities in need and how intelligent welfare systems can help Islamic organizations deliver Zakat to the needy Asnaf.

According to Zamzuri et al. (2020), ensemble machine learning techniques like random forest and XG Boost were utilized to predict poverty classifications through socioeconomic data sets. The research indicated that ensemble learning plays a crucial role in improving predictive accuracy in comparison to traditional statistical approaches. It shows the effectiveness of artificial intelligence in assisting poverty reduction efforts, including planning Islamic social welfare programs efficiently to sustain the welfare of poor Muslim families and Asnaf groups.

In a recent study by Sohnesen and Stender (2021), the effectiveness of machine learning algorithms in the prediction of poverty in developing nations was assessed with the use of Random Forest regressions. It was found that machine learning algorithms performed better than conventional statistical methods in predicting poverty levels because of their ability to accommodate complicated socio-economic interactions. These results point to the benefits of using artificial intelligence to enhance poverty monitoring and welfare planning, which will be useful to Islamic social finance organizations in facilitating Zakat distribution to impoverished Muslims.

Ishak and Jaapar (2020) utilised the Holt-Winters time series model to predict trends in Zakat collection, illustrating the efficacy of statistical forecasting in Islamic financial planning. Their findings indicate that Zakat payments may be forecasted with considerable precision, facilitating improved planning for welfare allocation among Asnaf. Nonetheless, the study exclusively examines financial inflows and does not extend to direct predictions of poverty levels, hence constraining its applicability in comprehensive poverty reduction programs.

Yusof et al. (2021) employed LSTM deep learning models to enhance the forecasting precision of Zakat collection over time. The research emphasises the power of artificial intelligence to identify intricate temporal trends in Islamic financial data. Despite its great accuracy, the model suffers from a lack of interpretability and necessitates extensive datasets, rendering it impractical for smaller Islamic institutions lacking modern computational resources.

Saharudin et al. (2023) examined the correlation between the macroeconomic conditions and poverty rate in Malaysia based on the Nonlinear Auto Regressive Distributed Lag approach. It was established that there were significant correlations between economic factors and the rate of poverty in the country. The poverty rate was highly dependent on macroeconomic stability. The study, however, is restricted to macro-analysis only.

Nurfazrina et al. (2023) employed ensemble machine learning models, including Random Forest and XGBoost, for the classification of poverty utilising socio-economic datasets. Their research indicates that machine learning markedly enhances the precision of poverty classification relative to conventional methods. The study depends on static statistics and fails to include real-time data or Islamic welfare metrics, hence constraining its relevance in dynamic poverty contexts.

The hybrid machine learning system was suggested by Ngong'ho Bujiku Sende et al. (2024) through an amalgamation of clustering and decision trees to predict poor households. This study has shown that a combination of both unsupervised and supervised learning systems enhances classification efficiency and minimizes human biases in identifying poor people. Nevertheless, this model is highly dependent on accurate structured data.

Aulia et al. (2020) investigated the application of alternative digital footprint data for poverty prediction employing K-Nearest Neighbour (KNN) and feature selection methodologies. The research demonstrates that unconventional data sources, such as digital engagement, may successfully categorise poverty levels. Nonetheless, its usefulness is constrained in rural or low-digital-access communities, which frequently represent the most vulnerable demographics in developing nations like Malaysia.

Finally, Ramzan et al. (2024) presented a deep learning system that employs gated-attention feature fusion and convolutional neural networks (CNN) for the prediction of poverty. The research exhibits superior predictive efficacy utilising geographical data, including satellite images. Nonetheless, it lacks integration with socio-economic and Islamic welfare metrics, such as Zakat and Asnaf classification, rendering it less appropriate for localised poverty study in Malaysia.

## IX. METHODOLOGY

### *A. Dataset Description*

This study is based on two datasets related to poverty and distribution of zakat in Malaysia. The poverty data was retrieved from the Department of Statistics Malaysia (DOSM) via the OpenDOSM platform, with the distribution of zakat data collected from Jabatan Wakaf, Zakat dan Haji (JAWHAR) via the official Baitulmal statistics portal. Both data sets include state level observations from 2007 up to 2022. The variables of the poverty dataset are Year, State, and Poverty rate, while the variables of the zakat dataset are Year, State, and Total Zakat Distribution. The datasets were merged based on the common attributes of Year and State to create a single dataset that includes poverty rate data and zakat distribution data across the states in Malaysia. This integrated data allows for analysis of poverty trends, welfare resource allocation and the correlation between poverty rates and the distribution of zakat, which forms the basis of the descriptive and predictive analysis undertaken in this study.

### *B. Data Analytics Tools*

This study was conducted using Python in the Google Colab, which provides a cloud-based platform for data analysis and machine learning. Several Python libraries were utilized throughout the project, including Pandas for data loading, cleaning, transformation, and merging; NumPy for numerical computations; Matplotlib for data visualization and exploratory data analysis; Scikit-learn for data splitting, predictive modelling, and model evaluation; and Statsmodels for statistical analysis of the Multiple Linear Regression model, including regression coefficients, t-values, and p-values. The data was preprocessed, analysed, visualised, modelled and evaluated in this study with the aid of these tools.

### *C. Data Collection and Preprocessing*

The data for this study were gathered from official government-related sources, with poverty rate data from the Department of Statistics Malaysia (DOSM) via the OpenDOSM platform and data on the distribution of zakat from the Jabatan Wakaf, Zakat dan Haji (JAWHAR) statistics portal. The data on zakat was extracted and structured using a web scraping method implemented in Python with the help of the libraries Requests, BeautifulSoup, and Pandas. Pre-processing methods were then applied to ensure data consistency and quality, such as standardizing variable names, converting state names to uppercase, converting date values to year format, and identifying missing values, duplicate records, and inconsistencies. Finally, both datasets were merged using the common attributes of Year and State, and the final unified dataset of poverty rate and zakat distribution information was obtained for the Malaysian states from 2007 to 2022 for further analysis and predictive modelling.

### *D. Exploratory Data Analysis (EDA)*

Exploratory Data Analysis (EDA) was conducted to understand the characteristics, trends, and relationships within the dataset before developing the predictive models. Data were summarized using descriptive statistics such as mean, minimum, maximum and standard deviation, while various visualizations were produced to highlight the pattern of poverty and zakat distribution in states in Malaysia. Bar charts were used to compare the poverty rates of different states, line charts were used to show trends over time between 2007 and 2022, and scatter plots were used to show the relationship between poverty rates and zakat distribution. The findings from this analysis were used to create a foundation for selecting the features to be used in predictive modelling and to ensure the data is appropriate for machine learning applications.

*E. Feature Selection*

To identify the variables relevant for predictive modelling, feature selection was carried out. The dataset is composed of four variables: Year, State, Poverty Rate, and Total Zakat Distribution. The poverty rate was chosen as the target and year and total zakat distribution as the predictor variables because of their relevance in capturing the temporal trends and the patterns of allocation of welfare resources related to the level of poverty. State variables were included in descriptive and exploratory analyses but were not included in the predictive models. The selected features were used as input features for the Multiple Linear Regression model and Decision Tree Regression model.

*F. Data Splitting*

The data was split into training and testing sets using an 80:20 ratio after feature selection, with 80% of the observations assigned to the training set for creating the predictive models and 20% assigned to the testing set for assessing the performance of the models on unseen data. Random sampling with a fixed random state value of 42 was used in the splitting process to ensure reproducibility and consistency of the results. The resulting datasets were then used for the development and evaluation of the Multiple Linear Regression model and the Decision Tree Regression.

*G. Predictive Modelling*

The two predictive modelling techniques are Multiple Linear Regression and Decision Tree Regression which were implemented to predict the Poverty Rate in Malaysia, where the Year and Total Zakat Distribution were used as the predictor variables and the Poverty Rate was used as the target variable. The models were chosen to test linear and non-linear prediction of poverty.

1) Multiple Linear Regression

Multiple Linear Regression was used to model the relationship between poverty rates and the selected predictor variables. The model estimates poverty rates based on the linear equation

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon$$

where:

- $Y$ represents the Poverty Rate,
- $X_1$ represents Year,
- $X_2$ represents Total Zakat Distribution,
- $\beta_0$ is the intercept,
- $\beta_1$ and $\beta_2$ are regression coefficients,
- $\varepsilon$ represents the error term.

The model was created with the use of the LinearRegression function from the Scikit-learn library, which was used to learn the patterns from the training dataset and make predictions for the testing dataset.

2) Decision Tree Regression

To handle the non-linearities between the variables, Decision Tree Regression was used as an alternative model. A model divides the data into smaller subsets with decision rules based on predictor variables and then makes predictions using a tree-like structure. The DecisionTreeRegressor function from the Scikit-learn library was used with a random state value of 42 to ensure reproducibility, allowing its predictive performance to be compared with that of Multiple Linear Regression.

*H. Model Evaluation*

The Multiple Linear Regression and Decision Tree Regression models were tested on the testing dataset, assessing the models based on the poverty rate predicted by the models and the actual poverty rate. The prediction accuracy and model performance was assessed using three regression metrics, namely Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Coefficient of Determination ($R^2$ Score). Furthermore, a statistical significance test using t-values and p-values was performed on the Multiple Linear Regression model; variables having p-values less than 0.05 were regarded as statistically significant. Scatter plots of actual versus predicted values were also created to visually compare the predictive ability of both models.

1) Mean Absolute Error (MAE)

The Mean Absolute Error (MAE) is the average of the absolute differences between the observed and predicted poverty rates. A smaller MAE results in a more accurate prediction.

$$MAE = (1/n) * \Sigma |y - \hat{y}|$$

where:

- n is the number of observations,
- $y$ is the actual poverty rate,
- $\hat{y}$ is the predicted poverty rate.

2) Root Mean Squared Error (RMSE)

Root Mean Squared Error (RMSE) is the square root of the average squared prediction errors, and larger

errors are penalized more heavily. Smaller RMSE values mean better model performance.

$$RMSE = \sqrt{[\,(1/n) * \Sigma\,(y - \hat{y})^2\,]}$$

where:

- n is the number of observations,
- $y$ is the actual poverty rate,
- $\hat{y}$ is the predicted poverty rate.

3) Coefficient of Determination ($R^2$ Score)

The Coefficient of Determination ($R^2$ Score) is a measure of the amount of variance in poverty rates accounted for by the model. Closer to 1 means a better ability to explain and predict. .

$$R^2 = 1 - [\,\Sigma(yi - \hat{y}i)^2 / \Sigma(yi - \bar{y}i)^2\,]$$

where:

- $yi$ is the actual poverty rate,
- $\hat{y}i$ is the predicted poverty rate,
- $\bar{y}i$ is the mean poverty rate.

I. Workflow Diagram

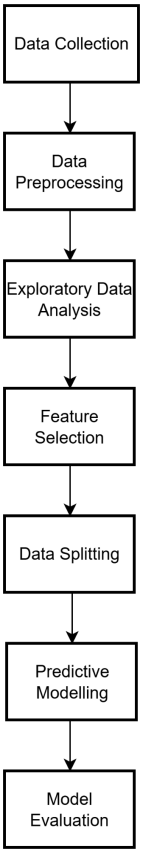



Figure 1. Workflow of methodology

## X. RESULTS

*A. Descriptive Analysis Result*

Figure 2: Average Poverty Rate by State

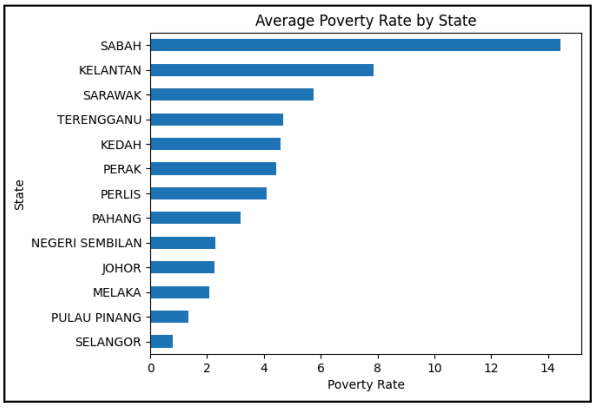



Figure 2 shows the average poverty rate across Malaysian states. Sabah recorded the highest average poverty rate (approximately 14%), followed by Kelantan and Sarawak. Selangor recorded the lowest average poverty rate.

Figure 3: Average Zakat Distribution by State

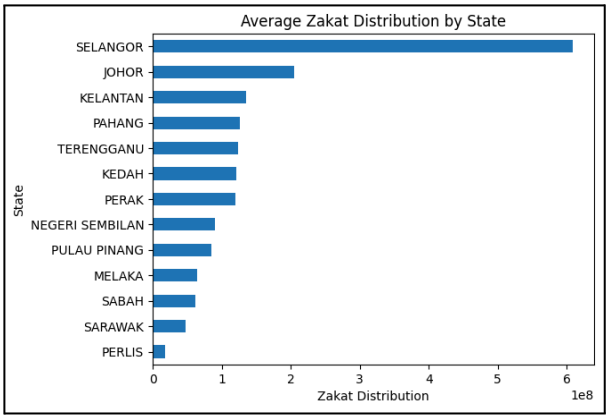



Figure 3 illustrates the average zakat distribution by state. Selangor received the highest average zakat distribution, followed by Johor. The remaining states exhibited relatively smaller distribution amounts.

Figure 4: Poverty Trend Over Years

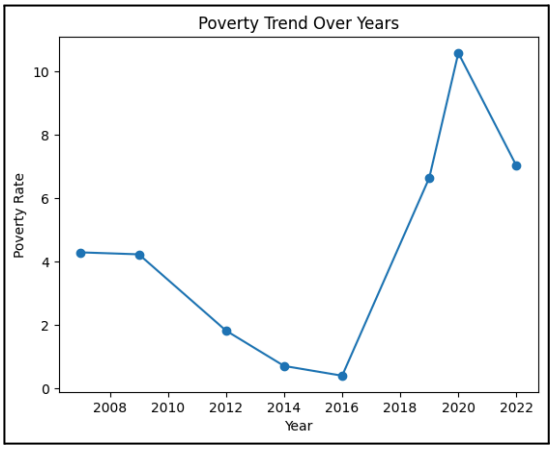


Figure 4 presents the average poverty rate from 2007 to 2022. The poverty rate generally decreased until 2016, increased sharply between 2018 and 2020, and subsequently declined after 2020.

Figure 5: Zakat Distribution Trend Over Years

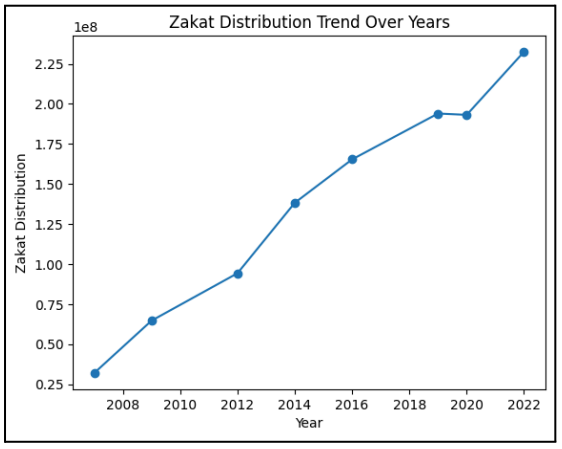



Figure 5 shows that zakat distribution generally increased throughout the study period, with slight fluctuations around 2019 and 2020.

Figure 6: Poverty Rate versus Zakat Distribution

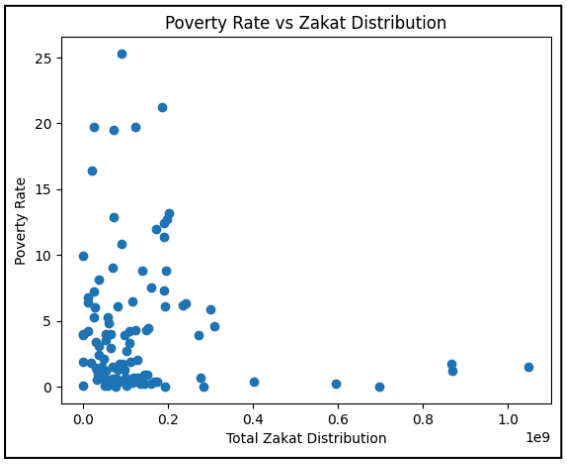



Figure 6 illustrates the relationship between poverty rate and zakat distribution. The distribution of points suggests a weak and non-linear pattern.

*B. Feature Selection and Correlation Results*

Table 1: Correlation Matrix

| Variable | Year | Zakat | Poverty |
|---|---|---|---|
| Year | 1 | 0.366 | 0.319 |
| Total Zakat Distribution (RM) | 0.366 | 1 | -0.111 |
| Poverty | 0.319 | -0.111 | 1 |

Table 1 presents the correlation coefficients among the selected variables.

*C. Multiple Linear Regression Results*

Table 2: Regression Coefficients

| Variable | t-value | p-value |
|---|---|---|
| Year | 4.150 | 0.000 |
| Total Zakat Distribution (RM) | -2.641 | 0.010 |

Table 2 shows the t-values and p-values obtained from the Multiple Linear Regression model. As both t-values are far from 0 and p-values are $< 0.05$, both variables are statistically significant.

*D. Model Performance Results*

Table 3: Performance Comparison

| Model | MAE | RMSE | $R^2$ |
|---|---|---|---|
| Multiple Linear Regression | 3.283 | 4.235 | 0.213 |
| Decision Tree Regression | 2.324 | 3.664 | 0.411 |

Table 3 presents the performance metrics of MAE, RMSE and $R^2$ for both models.

*E. Actual vs Predicted Correlation*

Table 4: Actual vs Predicted Correlation

| Model | Pearson Correlation | p-value |
|---|---|---|
| Multiple Linear Regression | 0.521 | 0.015 |
| Decision Tree Regression | 0.656 | 0.001 |

Table 4 shows the correlation between actual and predicted poverty rates for both models.

*F. Future Forecasting Results*

Figure 6: Future Poverty Prediction

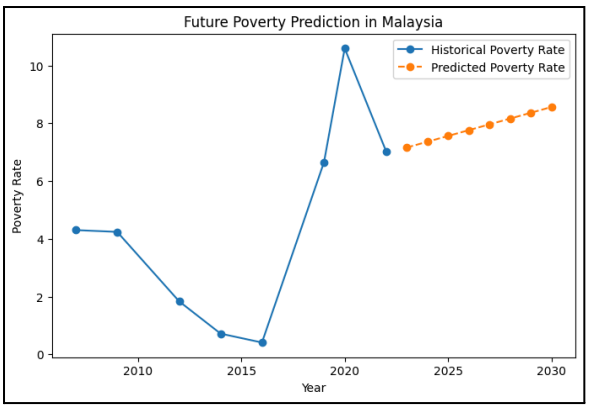



Figure 6 illustrates the predicted poverty rate from 2023 to 2030 generated using the Multiple Linear Regression model.

## XI. DISCUSSION

This section discusses the results of descriptive and predictive analytics gained from the analysis of poverty and Zakat distribution. It focuses on the correlation between poverty rate and Zakat distribution. the predictive capabilities of the machine learning models and the practical implications of the results for community welfare and poverty alleviation programmes in Malaysia.

### *A. Poverty Trends and Zakat Distribution*

There is a significant difference in the poverty rates among Malaysian states revealed from the descriptive analytics. As shown in Figure 1, Sabah recorded the highest average poverty rate (approximately 14%), followed by Kelantan and Sarawak, while Selangor recorded the lowest average poverty rate. These differences indicate persistent socio-economic challenges that may be influenced by factors such as employment opportunities, economic development, household income levels, and access to social support systems.

Similarly, variation in the allocation of welfare resources across states and years was observed through the analysis of Zakat distribution. Figure 2 shows that Selangor received the highest average Zakat distribution, followed by Johor, whereas the remaining states recorded considerably lower amounts. Generally, states with larger populations and greater welfare needs received higher levels of assistance. The visualizations created through line charts, bar charts and scatter plots suggested that Zakat institutions actively channel resources towards communities experiencing greater financial hardship.

Furthermore, Figure 5 illustrates the relationship between poverty rate and Zakat distribution. The scatter plot exhibits a weak and non-linear pattern, which is further supported by the correlation coefficient shown in Table 1, where the correlation between poverty rate and Zakat distribution was relatively weak (-0.111). The relationship between poverty and Zakat distribution highlights the importance of Islamic social finance in supporting vulnerable populations. Zakat alone is insufficient to eradicate poverty completely but it elevates the financial burden by providing temporary relief to eligible Asnaf. This supports the argument that welfare interventions should continue to be integrated with broader poverty reduction policies.

### *B. Forecasting Poverty Using Machine Learning Models*

Two predictive models, Linear Regression and Decision Tree Regression were developed to test whether machine learning could predict future trends in poverty and help plan welfare services in advance. A transparent and interpretable model was obtained by Linear Regression which explained the changes in poverty rates with respect to year and Zakat distribution. Table 2 shows that both predictor variables were statistically significant, with Year having a t-value of 4.150 ($p = 0.000$) and Zakat Distribution having a t-value of -2.641 ($p = 0.010$). Since both p-values are below 0.05, the predictors are considered statistically significant. Policymakers will be able to understand the impact of each predictor variable and to interpret the relationships mathematically through the model.

Decision Tree Regression provides more flexibility by recognizing non-linear correlations and hierarchical decision patterns. The model can capture more intricate interactions within the dataset because, in contrast to linear regression, it does not assume a fixed linear connection between variables. The Decision Tree model is particularly helpful in recognizing poverty patterns that are impacted by a variety of interrelated causes that don't follow a straightforward linear pattern.

An objective assessment of prediction accuracy is provided by comparing model performance using Mean Absolute Error, Root Mean Square Error, and $R^2$ values,as presented in Table 3. Multiple Linear Regression achieved an MAE of 3.283, RMSE of 4.235 and $R^2$ of 0.213, whereas Decision Tree Regression obtained lower prediction errors with an MAE of 2.324 and RMSE of 3.664, together with a higher $R^2$ value of 0.411. These results indicate that the Decision Tree model provided better predictive performance and explained a larger proportion of variation in poverty rates.

The correlation between actual and predicted values shown in Table 4 further supports this finding. Decision Tree Regression achieved a higher Pearson correlation coefficient (0.656, $p = 0.001$) compared to Multiple Linear Regression (0.521, $p = 0.015$), indicating stronger agreement between the predicted and actual

poverty rates. Therefore, machine learning techniques can serve as useful tools for supporting evidence-based welfare planning and forecasting socio-economic indicators.

Lastly, Figure 6 presents the projected poverty rates from 2023 to 2030 generated using the Multiple Linear Regression model. The forecasting results provide an estimate of future poverty trends that can assist policymakers and welfare institutions in planning resource allocation and intervention strategies.

*C. Practical Impact and Decision Support Applications*

This study demonstrates the usefulness of integrating descriptive and predictive analytics into welfare planning and poverty management, even beyond the forecasting outcomes. Poverty forecasting and decision support systems are achieved successfully through the project which may help stakeholders in decision making. Policymakers can utilize the forecasting results shown in Figure 6 to identify potential increases in poverty levels and implement preventive measures before conditions worsen. This supports more effective use of public funds and focused initiatives to reduce poverty.

By using forecasting, strategic financial support distribution for Zakat institutions and Islamic charity organizations can become better. Rather relying on historical trends, Institutions can anticipate future demand for welfare resources and develop suitable intervention methods by using forecasted poverty levels. The project's visualization component, represented by Figures 1 to 5, also functions as a monitoring tool, allowing stakeholders to determine long-term trends in poverty, evaluate welfare outcomes, and assess the effectiveness of existing poverty reduction initiatives. A data-driven strategy like this promotes increased accountability, efficiency, and openness in welfare management.

Overall, combining Zakat distribution analysis, poverty analysis and machine learning forecasting can support more proactive planning for community welfare. Malaysia's continuous efforts to achieve Sustainable Development Goal 1 (No Poverty) and enhance the socioeconomic well-being of vulnerable people are aided by the integration of descriptive and predictive analytics.

## XII. FUTURE WORKS

There is a great possibility for future studies in poverty prediction and forecast studies in Malaysia to enhance further the strength of the predictive models through the introduction of other variables in the analysis aside from the socio-economic ones. One particular study that could be considered as a contribution to this area of study is the development of an interactive dashboard that provides information on poverty trends and the distribution of zakat among different Malaysian states.

Apart from the inclusion of other socio-economic variables in the study, there is also the possibility of utilizing other socio-economic indicators such as household income, unemployment rates, education attainment, and demographics to further improve the accuracy of the prediction models.

Moreover, GIS-based spatial analysis can be applied to pinpoint the location of poverty hot spots within districts or even sub-districts. The advantage here is that resource allocations would be made in accordance with poverty concentration.

Other possible future directions include applying machine learning and deep learning approaches like Random Forest, XGBoost, and LSTM to improve prediction results since these algorithms can find non-linear relations between variables.

Finally, one can explore the integration of real-time or continuously updated datasets into the model to develop a dynamic forecasting tool. With this kind of dataset, one will be able to update forecast estimates in real time, resulting in more effective poverty reduction in line with SDG 1.

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